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# The calcium channels: one permeability, many origins

*A review of the pathways that carry $Ca^{2+}$ down its gradient through an open pore — their filters, their folds, their evolution, and what the sequence databases do and do not record about them.*

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## Summary

Calcium is the only ion whose channels are named for what they let through rather than for what they are. Potassium channels can be defined structurally: the T-x-G-Y-G filter is a sequence motif, and the proteins that carry it are the proteins that conduct $K^+$. Sodium channels are nearly as clean. **A calcium census has no such handle.** "Calcium channel" names a permeability ratio — a quantity measured in a particular cell, at a particular divalent concentration, sometimes per subunit combination, and in one important case per RNA-editing state of a single codon.

This review is organised around that fact and its consequences. The pores that conduct $Ca^{2+}$ have arisen independently in a large number of unrelated folds, so there is no alignment containing a ryanodine receptor and an ORAI channel and therefore no tree of calcium channels. What there is instead is a set of independent inventions that converged on the same physical problem: how to select a divalent ion that is thirty thousand times less abundant than sodium in the cytosol's own gradient, and still conduct it at a rate near diffusion.

We take the solutions one at a time — the four-repeat voltage-gated pore, the CRAC channel's single glutamate ring, the uniporter's five-residue filter, the epithelial TRPV aspartates, the intracellular release channels, the ligand-gated receptors, the mechanosensors, and the plant, fungal and viral channels that are not the animal ones renamed — and ask of each what is genuinely shared and what merely looks shared.

Two questions then have to be answered before anything can be counted. The first is where the boundary of the subject falls, because the word "calcium" attaches to $Ca^{2+}$-*activated* channels that conduct $K^+$ or $Cl^-$, to pumps and exchangers that are not channels, to EF-hand sensors that conduct nothing, and to sixteen human genes whose symbols begin `CACN` and which encode no pore at all. The second is what the

annotation databases actually record, which turns out to be less than a census needs and to fail in ways that are systematic rather than random.

## Scope, and what "the census" means here

Every count in this review is a count under the boundary decisions recorded in `docs/scope_and_boundaries.md`, and is reported in tiers rather than as a single headline number. The catalogue behind it holds **245 human genes and counts 98**: 28 in the selective tier, 50 permeable, 9 conditional and 11 release. The 147 it does not count are catalogued too, each with a recorded reason and, wherever a test exists, a positive test identifying it as something else. That ratio is the point. A review that listed only what it counted would look tidier and would be impossible to check.

## Figures

| | |
|---|---|
| **1** | the folds that build a calcium-permeable pore, one subunit each — *schematic* |
| **2** | the potassium filter across the P-loop branch — *measured, no alignment used* |
| **3** | the four-repeat locus: EEEE / EEDD / DEKA / EEKE — *measured, projected from Nav1.5* |
| **4** | length range by superfamily — *measured* |
| **5** | which families carry which shared domain, and which cross the census boundary — *measured* |
| **6** | a rooted ML tree of the intracellular release channels — *not yet rendered; awaits the pore-module extraction of S6 and the tier-2 build of S8* |
| **7** | the forest, and the superfamilies that get no tree — *schematic of a rule* |
| **8** | the architecture traps, to scale — *measured* |

Seven of the eight are rendered; figures rendered from committed tables carry the table they came from in their legend, and the schematics say so and display no data beyond the counts in them. **Figure 6 is deliberately absent.** A tier-2 tree requires the pore module to be extracted first, because a full-length alignment of a 5,038-residue ryanodine receptor against a 2,700-residue $IP_3$ receptor is dominated by the cytoplasmic caps rather than by the pore. Running the command on full-length sequences produces a tree, and the tooling itself reports that the result is a quick look rather than the tier-2 result. Substituting it here would be the kind of quiet methodological downgrade this project forbids, so the figure waits for S6 and S8.

## Key points

- **Calcium-channel membership is a permeability, not a clade.** There is no motif and no fold that defines the subject, so the census is reported in four evidence tiers and a flat total is never quoted without them.
- **The nearest relative of a calcium channel is usually not one.** Cav sits beside Nav and is separated from it by four residues; CNG beside HCN; TRPV6 beside TRPV1; TRPM7 beside TRPM4. Domain architecture cannot make any of these calls.
- **$Ca^{2+}$ selectivity has been invented repeatedly in unrelated folds.** The four best-characterised solutions — the EEEE ring, ORAI's E106, the uniporter's DIME motif and the TRPV aspartate ring — share no alignable position with one another.
- **Some permeabilities are not in the genome.** Whether an AMPA receptor conducts $Ca^{2+}$ is decided by ADAR2 editing of a single GRIA2 codon, and no sequence-level method can recover it.
- **The word "calcium" is the census's largest error source.** The $Ca^{2+}$-activated channels conduct $K^+$ or $Cl^-$; the pumps and exchangers are transporters; the EF-hand sensors conduct nothing; and of the sixteen `CACN`-prefixed human genes that encode no pore, seven belong to the AMPA receptor.
- **Domain annotation does not partition the subject the way the textbooks do**, and a census built on the obvious accessions loses whole divisions without saying so.

## 1. Why calcium, and why it is hard to count

### 1.1 An ion chosen for its scarcity

Calcium signalling begins with an accident of chemistry that life turned into an advantage. Ringer found in 1883 that a frog heart would not keep beating in saline made without calcium, and that restoring it restored contraction [1]. Seventy years later Fatt and Katz recorded, in crustacean muscle, an action potential that persisted when sodium was removed and that was carried instead by calcium — the first evidence of a voltage-gated calcium conductance [2]. The two observations bracket what makes the ion useful: it is required, and it is separately controllable.

The reason is a gradient maintained at metabolic cost. Cytosolic free $Ca^{2+}$ rests near 100 nM against roughly 2 mM outside and 0.3–0.8 mM in the endoplasmic reticulum, a ratio of about ten thousand to one — steeper than any other physiological ion gradient. A cell that opens a calcium-permeable pore therefore gets a large, fast, local signal from a very small number of ions, and one that can be terminated by buffering and pumping rather than by bulk transport. Calcium is a second messenger because it is scarce, and the machinery that keeps it scarce is what makes the message legible [3,4,5,6].

The techniques that made these currents legible were developed on other ions and inherited. The Hodgkin–Huxley description of the squid axon established that a membrane current could be decomposed into independent, voltage- dependent conductances and modelled quantitatively before any channel protein was known [7], and single-channel recording made the discrete conducting unit directly visible [8]. Both were applied to calcium currents almost immediately, and the vocabulary of this review — selectivity ratios, activation thresholds, single-channel conductance — is theirs.

That scarcity sets the engineering problem the rest of this review is about. A calcium-selective pore must discriminate $Ca^{2+}$ from $Na^{+}$ and $K^{+}$ that are present at a hundred to a thousand times its concentration, and must do so while conducting at rates approaching free diffusion. High selectivity normally means tight binding, and tight binding normally means slow release. The resolution — a pore that binds calcium tightly enough to exclude monovalents but holds more than one ion at a time, so that electrostatic repulsion between them restores a high throughput rate — was worked out for the L-type channel and remains the general principle [9,10]. Nearly every solution described below is a variation on it.

### 1.2 A functional class, not a clade

The term “calcium channel” was never a phylogenetic claim. It describes behaviour, and the behaviour has been arrived at many times. This review’s catalogue places the pore-forming calcium channels in **17 superfamilies**, several of which share no alignable position with any other: the P-loop channels, the TRP division, the intracellular release channels, ORAI, the mitochondrial uniporter, VDAC, the Cys-loop receptors, the ionotropic glutamate receptors, P2X, Piezo, the OSCA/TMEM63 mechanosensors, the CALHM large pores and the viral viroporins, among others.

This has a consequence that organises the entire project behind this review: **there is no tree of calcium channels.** An alignment column that contains a residue of a ryanodine receptor and a residue of ORAI1 does not exist, and a tree built across them would be a picture of an alignment artefact. What can be built is a *forest* — trees within families and within superfamilies, where descent is real — plus a structural network between the folds, which carries no branch lengths and no support values because it is not a phylogeny. Any sentence beginning “calcium channels evolved…” is therefore either about one superfamily or is wrong.

Three long-standing observations make the same point from the other direction, and each is a place where the obvious grouping misleads:

- **Cav and Nav are the same architecture.** Both are single polypeptides of four homologous six-transmembrane repeats surrounding one pore. They are separated by the four residues that each repeat contributes to the selectivity filter — EEEE in Cav1/Cav2 against DEKA in Nav — and a single substitution in the sodium channel confers calcium selectivity on it [11].
- **Calcium permeability varies within families.** TRPV5 and TRPV6 are $Ca^{2+}$-selective and their close relatives TRPV1–TRPV4 are not; the nicotinic α7 subunit reaches P_Ca/P_Na near 10 while its sister α4 is near 1 [12,13].
- **Calcium permeability is sometimes not encoded at all.** An AMPA receptor containing an edited GluA2 subunit is calcium-impermeable; the same gene, unedited, is not. The decision is made post-transcriptionally by ADAR2 on a single codon [14,15].

### 1.3 A definition, and its five edges

Throughout this review a calcium channel is:

> a protein assembly forming a **gated aqueous pore** that conducts $Ca^{2+}$ across a membrane **down its electrochemical gradient**, without a coupled conformational transport cycle.

Each clause marks a boundary that published gene lists draw differently, and each is worth stating because the disagreements account for most of the variation between counts.

1. **Selectivity is not required, but it is graded.** Demanding P_Ca/P_Na > 10 would exclude most TRP channels, every $Ca^{2+}$-permeable AMPA receptor, P2X, Piezo and CNG — that is, most of the calcium entry in most non-excitable cells. Admitting anything with a measurable $Ca^{2+}$ current would admit the muscle nicotinic receptor and, in the limit, any large pore. This review therefore reports four tiers with the evidence for each attached, and never a single number.
2. **Gating *by* calcium is not membership.** BK and SK channels, TMEM16A and TRPM4/TRPM5 are opened by cytosolic $Ca^{2+}$ and conduct $K^+$, $Cl^-$ or monovalent cations [16,17]. They carry "calcium" in their names and in most review titles, and they are the single most common contaminant of keyword-built lists.
3. **Transport is not conduction.** SERCA, PMCA and SPCA hydrolyse ATP; NCX, NCKX and NCLX are coupled to the sodium gradient [18,19,20]. They move more calcium per cell than most channels do, and they are excluded on mechanism, not importance.
4. **Auxiliary subunits are not pores.** The Cav β, α2δ and γ subunits, STIM1 and EMRE are each required for a calcium channel to work and none of them lines a pore [21]. They are the largest single source of inflation in published totals.
5. **Release from a store counts, and is recorded as such.** $IP_3$ receptors, ryanodine receptors, TRPML and the two-pore channels do not admit calcium to the cell; they move it from an internal store into the cytosol, which is often the larger signal [22,23].

### 1.4 What follows

Sections 2–10 take the pores themselves: first the selectivity filters, because they are what the census ultimately turns on, then the families grouped by where their calcium comes from — the plasma membrane, the internal stores, the mitochondrion — and by what opens them. Section 11 works through the boundary explicitly, and is where the excluded set is argued rather than assumed. Section 12 reports what the annotation databases record, measured against live services rather than recalled. Sections 13–15 turn to evolution, disease and what remains unresolved, and section 16 states the methods.

## 2. Four filters, no common ancestor

If calcium channels formed a clade, they would share a filter. They do not. The four best-characterised calcium-selective pores are built from four different structural ideas, occupy four unrelated folds, and share no alignable position with one another. This section takes them in turn, because everything the census does downstream — the positive tests, the hazard rules, the refusal to build one tree — follows from the fact that they are separate inventions.

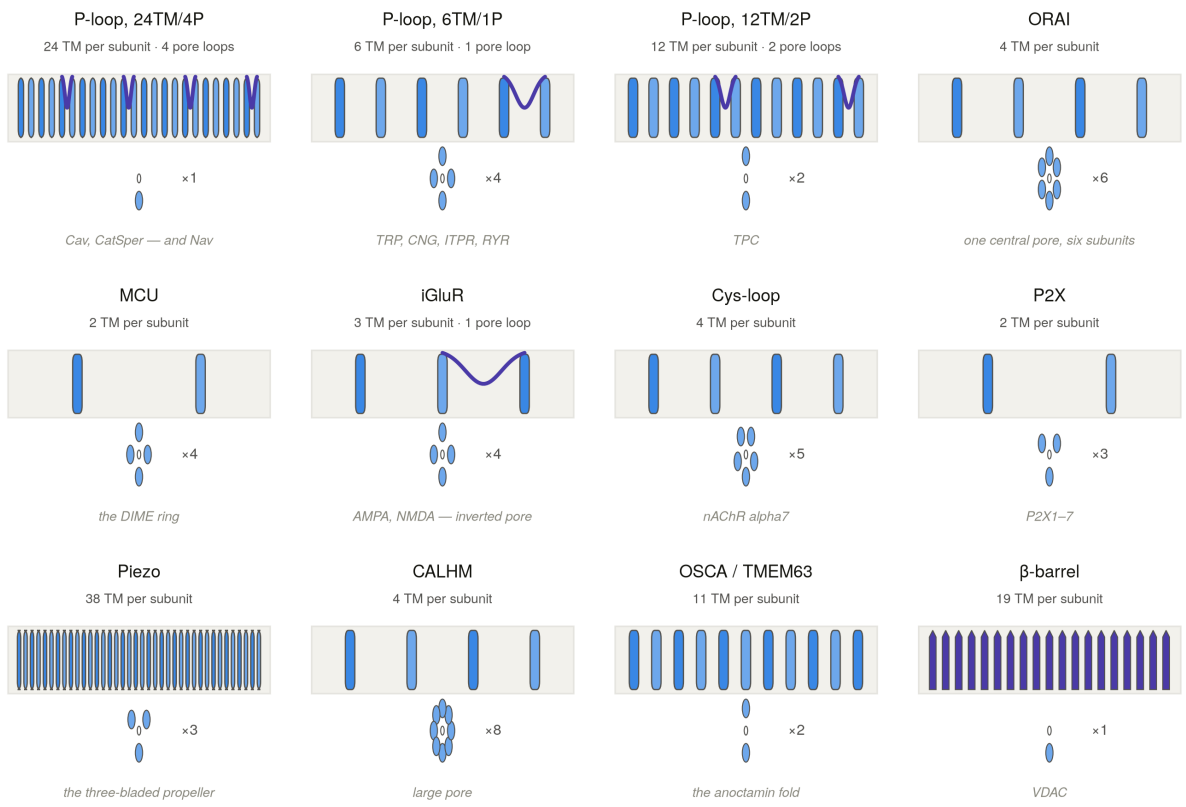



**Figure 1 | The folds that build a calcium-permeable pore.** One subunit each, drawn to a common scheme, for the folds represented in this review's catalogue. Transmembrane helices are boxes and pore-forming re-entrant loops are marked; the drawing is schematic and carries no data beyond the topology and the counts shown. The point of the figure is the absence of a common ancestor: no two of these can be aligned to one another, and the calcium selectivity of each was arrived at independently.

### 2.1 The EEEE ring: four repeats, four glutamates

The voltage-gated calcium channel is a single polypeptide of four homologous repeats, each contributing one re-entrant P-loop to a single central pore. Mutagenesis of the L-type channel identified one glutamate per repeat as the determinant of calcium selectivity, and neutralising them degrades selectivity in proportion to the number changed: the filter is a ring of four carboxylates, and it acts as a unit [10]. The complementary experiment is the more striking one. Substituting the corresponding residues of a sodium channel — whose own ring reads DEKA — converts it into a calcium-selective channel [11]. Four side chains separate the two largest families of voltage-gated cation channels in the human genome.

Structures later showed why the ring can be both selective and fast. The bacterial channel CavAb, engineered from a sodium-channel scaffold, and the rabbit Cav1.1 complex both resolve multiple $Ca^{2+}$ ions bound in file within the filter [24,25,26]. Selectivity comes from a high-affinity site that monovalents cannot compete for; throughput comes from a second ion arriving and repelling the first. Tight binding and fast conduction are reconciled by never having only one ion present.

The same locus separates four families that domain architecture cannot tell apart at all, and this review's project reads it as a four-residue string projected from human Nav1.5:

| filter | family | note |
|---|---|---|
| EEEE | Cav1, Cav2 | the high-voltage-activated locus |
| EEDD | Cav3 | the T-type locus |
| DEKA | Nav | a sodium channel |
| EEKE | NALCN | a sodium-leak channel [27] |

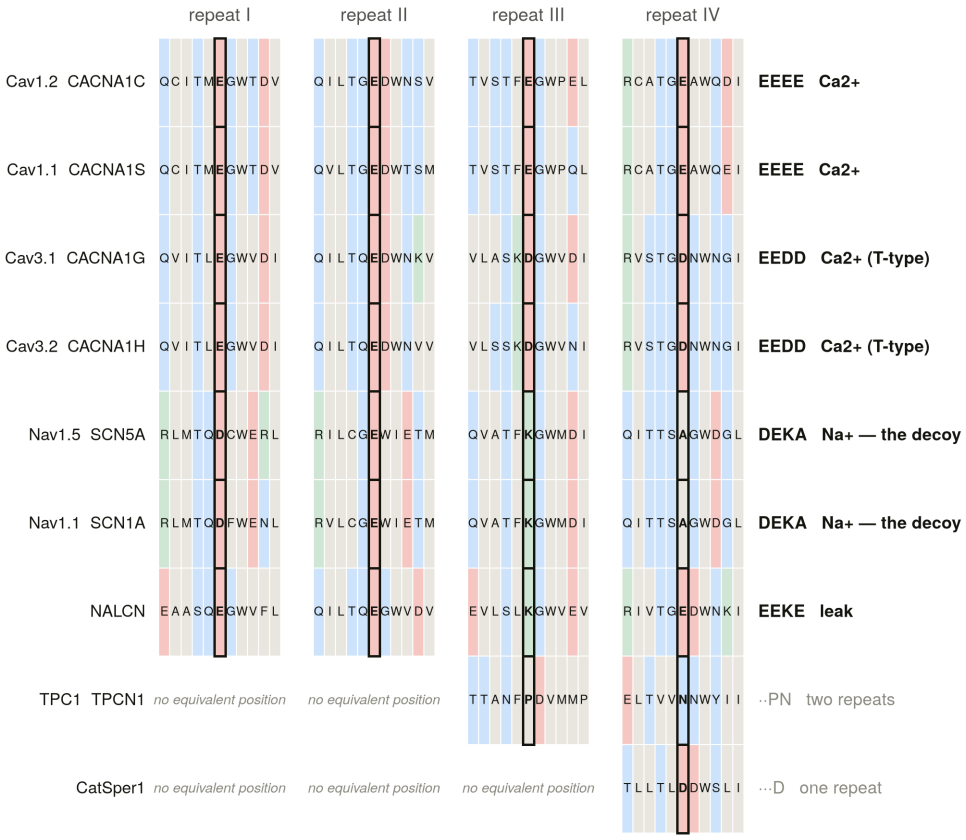



**Figure 3 | Four residues separate two superfamilies.** The four-repeat selectivity locus, projected from human Nav1.5 `Q14524` onto each query by alignment and read directly — rendered from `filter_four_repeat.tsv`. The high-voltage-activated calcium channels read EEEE, the T-type channels EEDD, the sodium channels DEKA and NALCN EEKE. Two rows carry the caution rather than the result: TPC1 returns a partial signature because it has two repeats rather than four, and CatSper1 returns a partial signature because its four positions lie in four different proteins. A partial result is recorded as partial, not discarded.

Cav3's EEDD is the reason the census cannot use "EEEE" as its test: the T-type channels are calcium channels with a different ring [28,29]. CatSper, the sperm channel, shares the same architecture again and cannot be read this way at all, because its four pore-forming subunits are four separate genes contributing one residue each [30,31].

### 2.2 ORAI: one glutamate, and a pore that is not a P-loop

The store-operated calcium current is the most calcium-selective conductance in the plasma membrane, with P_Ca/P_Na above 1000, and it is carried by a protein with no relationship to any voltage-gated channel. ORAI1 has four transmembrane helices, no P-loop, and assembles as a hexamer; its filter is a single glutamate, E106, contributed once per subunit [32,33,34,35]. The *Drosophila* Orai structure showed the ring directly, at the extracellular mouth of a long pore lined by basic residues [36]. Mutating E106 to glutamine abolishes calcium selectivity outright.

Structurally this is the same trick as the EEEE ring — a ring of carboxylates at the narrowest point — implemented in a fold that has nothing in common with the P-loop channels. It is the clearest single case of convergence in the subject.

### 2.3 The uniporter: the shortest filter, the highest selectivity

Mitochondrial calcium uptake was measured as a current in 2004 and shown to be the most calcium-selective channel then known, with an estimated P_Ca/P_Na above $10^4$ [37]. The protein responsible was

not identified for another seven years, by two groups in the same year, using comparative genomics built on the observation that yeast takes up no mitochondrial calcium and lacks the gene [38,39,40].

MCU's filter is a five-residue motif, W-D-x-x-E, known as DIME and sitting in a short loop between two transmembrane helices. Cryo-EM structures of fungal and metazoan uniporters resolve the two acidic rings it forms [41]. It is the only family in this review's catalogue that can be called from a bare regular expression on an unannotated sequence, and it achieves the highest selectivity of any of them with the smallest filter.

The uniporter also supplies the subject's cleanest boundary case. EMRE is required for conduction in metazoans and lines none of the pore [42]; MICU1 is an EF-hand gatekeeper that holds the channel shut at resting calcium [40]. Both are excluded from the census. MCUB, by contrast, assembles into the tetramer, contributes pore lining, and *reduces* conduction — and is counted, because the question the catalogue asks is whether a gene contributes pore lining, not whether it increases current.

### 2.4 The TRPV aspartate ring, and an anchor that had moved

TRPV5 and TRPV6 are the epithelial calcium channels of intestinal and renal reabsorption, with P_Ca/P_Na above 100 [43,44,45]. Their close relatives TRPV1–TRPV4 are non-selective cation channels. The discriminator is a ring of one aspartate per subunit in the outer pore, resolved in the TRPV6 structure [46].

This filter is the one that cannot be found by pattern, because the residue sits in no conserved context; it has to be projected from a reference sequence. Doing so exposed a trap worth recording, because it is the kind that fails silently. The literature numbers the TRPV6 aspartate D542. The canonical human UniProt entry `Q9H1D0` is the 765-residue long isoform and carries the filter `TIIDGP` at positions 579–584, putting the aspartate at **D582** — a 40-residue offset. An anchor taken from a paper's numbering would have aligned correctly, projected the wrong column, and returned a confident answer about the wrong residue. Every projection anchor in this project is therefore verified against its own reference before use, and a non-validating anchor refuses to project rather than guessing. All three anchors were re-verified against live UniProt on 2026-08-20: Nav1.5 `Q14524` reads DEKA at 372/898/1419/1711, ORAI1 `Q96D31` reads E at 106, and TRPV6 reads D at 582.

A caution belongs with this one. TRPV1–TRPV4 also carry acidic residues in the outer pore, so a projected aspartate is supporting evidence rather than a family call, and how far the anchor actually separates the two groups is a measurement the classifier benchmark makes rather than an assumption (*pending: S1*).

### 2.5 The negative test used positively

One motif does more work in a calcium census than any of the above: the potassium filter, T-x-G-Y-G. A protein carrying it conducts $K^+$, which settles the $Ca^{2+}$-*activated* potassium channels — BK, SK — without consulting a name, a symbol or an annotation. In a subject where the most common error is admitting something because it has "calcium" in its title, a reference-free regular expression that says "this conducts potassium" is the most reliable exclusion available.

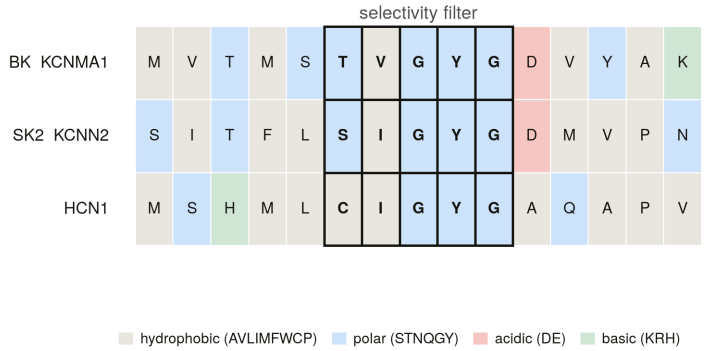



**Figure 2 | The exclusion that needs no reference.** The T-x-G-Y-G potassium filter located directly in each sequence with flanking context — no alignment was performed — rendered from `filter_k.tsv`. In a calcium census this is a negative test used positively: a protein carrying it conducts $K^+$, which removes the $Ca^{2+}$-activated potassium channels from the count without consulting a name, a symbol or an annotation.

The limits are equally worth stating. TRPM4 and TRPM5 are $Ca^{2+}$-activated and conduct monovalent cations rather than $K^+$, so no potassium filter identifies them; they share every Pfam accession and the same length

band with the $Ca^{2+}$-permeable TRPMs, and nothing in their sequence separates them. That split is a literature call, recorded as one, and the classifier is not expected to make it.

## 3. The voltage-gated calcium channels

### 3.1 Ten genes, three subfamilies, one architecture

The voltage-gated calcium channels are the family the term "calcium channel" usually means, and the one whose pharmacology reaches furthest into clinical practice. Ten human $\alpha_1$ genes encode the pore, and they fall into three subfamilies by sequence and by physiology [47,48]:

| subfamily | genes | current | activation |
|---|---|---|---|
| Cav1 | `CACNA1S`, `CACNA1C`, `CACNA1D`, `CACNA1F` | L-type | high-voltage |
| Cav2 | `CACNA1A`, `CACNA1B`, `CACNA1E` | P/Q-, N-, R-type | high-voltage |
| Cav3 | `CACNA1G`, `CACNA1H`, `CACNA1I` | T-type | low-voltage |

The purification that started the molecular era used the dihydropyridine receptor of skeletal muscle, and the sequence that followed established the four-repeat architecture [49]; the cardiac L-type channel followed [50], and the T-type channels last, a decade later [28,29]. Structures of the Cav1.1 complex resolved the pore, the four voltage sensors and the auxiliary subunits together [25,26].

Physiologically the split that matters is not Cav1 against Cav2 but high-voltage against low-voltage. Cav1 and Cav2 open on strong depolarisation and carry the calcium that triggers contraction and transmitter release; Cav3 opens near rest, inactivates quickly, and supports rhythmic firing and rebound bursts [28]. Cav1.1 in skeletal muscle is the extreme case: it works largely as a voltage sensor mechanically coupled to RyR1 rather than as a conduit, and calcium entry through it is not what triggers contraction [51].

### 3.2 The nearest relative is a sodium channel

Cav's closest architectural neighbours are not other calcium channels. The four-repeat P-loop channels comprise the voltage-gated calcium channels, the voltage-gated sodium channels, the sodium-leak channel NALCN and the sperm channel CatSper, and they are indistinguishable by domain architecture [27,52,53]. This project's own measurement makes the point concretely: `PF00520` in four copies describes all of them equally well, and CACNA1G carries `PF00520`×4 and nothing else — no Cav-specific domain at all.

The consequence for any census is direct. A calcium channel here cannot be recognised by what domains it carries; it has to be recognised by reading the filter (§2.1). This is the single most important hazard in the subject, and the reason the classification behind this review assigns by positive test rather than by architecture.

A second, subtler consequence concerns trees. Because superfamilies are defined by descent rather than by ion, a pore-module tree of the P-loop superfamily *contains* the sodium channels and the $Ca^{2+}$-activated potassium channels. Dropping them to produce "the calcium channel tree" would hide the non-monophyly and remove the outgroup that makes the tree rootable. The protocol behind this review forbids it.

### 3.3 The subunits that are not channels

The high-voltage-activated channels are complexes: an $\alpha_1$ pore with a cytoplasmic β subunit, an extracellular-facing α2δ and, in some tissues, a γ subunit [21,54]. None of them forms a pore, and all of them carry `CACN` symbols. Sixteen human genes do — four `CACNB`, four `CACNA2D`, eight `CACNG` — and this review's catalogue counts none of them.

Three facts make this more than nomenclature:

- **The β subunit is required for the channel to reach the membrane at all**, and shifts its voltage dependence by tens of millivolts. It is a MAGUK — a scaffold fold doing a channel job.
- **α2δ-1 is the gabapentin and pregabalin target.** An auxiliary subunit that conducts nothing is among the most-prescribed drug targets in the entire catalogue [55,56].

- **Seven of the eight `CACNG` genes are not calcium-channel subunits at all.** CACNG2–CACNG8 are TARPs: claudin-fold proteins that traffic and modulate AMPA receptors, with no role in voltage-gated calcium channels. Their symbols are a historical accident of the order of discovery, and they are counted as calcium-channel genes by any list built on symbol prefixes.

This project measured the architectures directly and found that the three subunit types share no domain at all: `CACNB1` carries a guanylate-kinase domain and a β-specific N-terminal domain, `CACNA2D1` a von Willebrand A domain and the α2δ domain, `CACNG2` a claudin domain alone. The "family" is three unrelated protein types grouped by the job they do.

### 3.4 CatSper, and a filter that cannot be read

CatSper is the principal calcium channel of sperm, required for hyperactivated motility and for male fertility, and gated in human sperm by progesterone acting through a lipid hydrolase rather than through a classical receptor [30,57,58]. It is a four-repeat channel built the hard way: four distinct pore-forming genes, `CATSPER1`–`CATSPER4`, each contributing one repeat, alongside several accessory subunits [31].

That construction defeats the test that works for every other four-repeat family. A filter projection reads four positions from one sequence; CatSper's four positions are in four different proteins, so no single-sequence projection can recover the ring. The locus is recorded as unread rather than guessed, and closing it would need a joint projection across all four subunits (*pending: S9*).

## 4. Release from internal stores

Calcium entering from outside is only part of the signal, and often the smaller part. The endoplasmic and sarcoplasmic reticulum hold free $Ca^{2+}$ at 0.3–0.8 mM against a cytosol at 100 nM, and the channels that open onto that gradient produce most of the calcium transients in most cells. They are counted here in their own tier, with the compartment recorded, because the same permeability ratio means something different on an organelle: what matters is not how well $IP_3R$ discriminates $Ca^{2+}$ from $K^+$ but the size of the gradient it opens onto.

### 4.1 $IP_3$ receptors and ryanodine receptors

The two large release channels are the largest proteins in this review's catalogue — RYR1 is 5,038 residues — and they are relatives. Both were cloned in 1989: the ryanodine receptor from skeletal muscle [59] and the $IP_3$ receptor as the cerebellar P400 protein [60], with the similarity between them noticed immediately [61]. Both are homotetramers with a small C-terminal pore module of the P-loop kind and an enormous cytoplasmic cap that acts as a signal-integration platform, resolved by cryo-EM in both families [62,63,64].

Their physiology is organised by what opens them. $IP_3$ receptors are gated by $IP_3$ produced downstream of phospholipase C, with cytosolic $Ca^{2+}$ as a co-agonist, giving the biphasic dependence that underlies calcium waves and oscillations [22]. Ryanodine receptors are gated by calcium itself — calcium-induced calcium release, demonstrated in skinned cardiac fibres long before the protein was known [23] — except in skeletal muscle, where RyR1 is opened by direct mechanical coupling to Cav1.1 [51,65].

For a census, the pair are a domain-sharing hazard rather than a discrimination problem: they share all four of `PF08709`, `PF01365`, `PF08454` and `PF02815`, so architecture places a sequence in the pair reliably and inside it not at all. One of those domains, the MIR domain `PF02815`, is also carried by the O-mannosyltransferase POMT1, which is not a channel — an illustration of why a domain hit is evidence about a superfamily and not about a function.

### 4.2 Two-pore channels: counted, and contested

TPC1 and TPC2 sit in endosomal and lysosomal membranes and are structurally the intermediate the four-repeat story predicts: two six-transmembrane repeats rather than four, a plausible waypoint between a 6TM ancestor and the duplicated Cav/Nav architecture.

Their permeability is genuinely disputed. They were identified as the NAADP target releasing calcium from acidic stores [66], and direct endolysosomal patch clamp subsequently found a $PI(3,5)P_2$-activated, $Na^+$-selective current instead [67]. This review counts them, in the release tier, and records the dispute rather than resolving it by omission; removing them costs the census two genes, and the catalogue is built so that this is a one-line change and a recount.

### 4.3 TRPML and the lysosome

The mucolipins TRPML1–3 (MCOLN1–MCOLN3) are the lysosome's own release channels, and TRPML1 loss causes mucolipidosis type IV. Their selectivity is measured at lysosomal pH, near 4.6, which is part of why their reported P_Ca/P_Na spans a wide range, and TRPML1 conducts $Fe^{2+}$ as well [68,69]. They belong to the TRP division by descent (§7) and are placed in the release tier by compartment.

### 4.4 The regulators that are not channels

Around both large release channels sits a set of proteins that are required for release and conduct nothing, and they are a good test of whether a gene list is counting pores or counting participants. Calsequestrin is a luminal calcium buffer whose occupancy sets how much RyR2 releases, and CASQ2 mutations cause catecholaminergic polymorphic ventricular tachycardia — the same disease as RYR2 mutations. FKBP12 and FKBP12.6 are immunophilins bound to every RyR subunit. Triadin and junctin link the two.

The catalogue records these as channel-associated, and this project's own architecture measurement found that they share no domain with one another: CASQ2 carries the calsequestrin domain alone, FKBP1A the FKBP domain alone. A buffer and an immunophilin have in common only the channel they regulate. TRIC channels (TMEM38A, TMEM38B) belong to the same category from the other direction: they are potassium channels that provide the counter-current for calcium release [70], and TMEM38B mutations cause a calcium-release disease through a potassium channel.

## 5. Store-operated entry: a gate in another organelle

Store-operated calcium entry is the mechanism by which emptying the ER opens a channel in the plasma membrane, and it was a physiological phenomenon for two decades before either of its two proteins was known. The current — CRAC, calcium release-activated calcium — is tiny, inwardly rectifying and extraordinarily selective, with P_Ca/P_Na above 1000 [71,72].

Its molecular basis arrived from two directions within about a year. RNAi screens in *Drosophila* identified STIM as the ER sensor [73,74], and then a genome-wide screen, a linkage study in a patient with severe combined immunodeficiency, and a parallel screen converged on ORAI1 as the pore [34,35,75]. That the same gene came out of an unbiased screen and a human immunodeficiency pedigree is a large part of why the assignment was accepted quickly. Pore-mutation experiments then established that ORAI1 is the channel rather than a regulator of one, by showing that altering E106 alters the selectivity of the current itself [32,33].

### 5.1 Why this family is architecturally isolated

ORAI has four transmembrane helices, no P-loop, no voltage sensor, and no detectable homology to any other channel family in this review. The hexameric *Drosophila* Orai structure showed a single long pore with the E106 ring at its extracellular mouth [36]. In the catalogue it is one family in its own superfamily and its own fold, and in the fold network it is one of the clearest examples of a $Ca^{2+}$-selective pore that owes nothing to the others.

For the census this makes ORAI easy in one respect and hard in another. It is easy to test — a single projected glutamate, verified against live UniProt at Q96D31 position 106 — and impossible to place in any tree containing another calcium channel.

### 5.2 STIM: required, and not a pore

STIM1 spans the ER membrane with a luminal EF-hand that senses store depletion; on emptying it oligomerises, migrates to ER–plasma membrane junctions and traps and opens ORAI directly. It is as necessary to the current as ORAI is, and it conducts nothing. The catalogue records STIM1 and STIM2 as channel-associated.

STIM also illustrates why an EF-hand cannot be used as evidence against channel identity. This project's canonical EF-hand detector finds four loops in calmodulin, four in calbindin, two in parvalbumin, two in MICU1 and one in STIM1 — and zero in CACNA1C and zero in MCU. EF-hands mark the calcium *sensors*, including two that sit inside census complexes as gatekeepers, and none of the pores. The motif is a useful positive detector for the decoys and never a veto.

## 6. The mitochondrial pathway

### 6.1 Fifty years between a current and a gene

Mitochondria take up calcium avidly, and the fact has been known since the 1960s. Direct patch clamp of the inner membrane finally measured the conductance responsible and showed it to be a channel of extraordinary selectivity, with an estimated P_Ca/P_Na above $10^4$ — greater than any plasma-membrane calcium channel [37]. The gene took another seven years.

The reason it took so long is instructive, and it is an evolutionary fact rather than a technical one. *Saccharomyces cerevisiae* has no mitochondrial calcium uptake, so the standard genetics of the era had no assay; and the protein has no homology to any known channel family, so sequence-based searching had no query. What eventually worked was comparative genomics across organisms that do and do not take up mitochondrial calcium: MICU1 came out of a phylogenetic-profiling screen [40], and MCU itself was identified the next year by two groups [38,39]. Structures followed [41].

MCU's filter is the `WDxxE` DIME motif (§2.3). The catalogue counts MCU and MCUB — the latter a pore-lining subunit that reduces conduction — and excludes MICU1–3, EMRE and MCUR1 as regulators [42].

### 6.2 VDAC, and a boundary this review draws loosely

The outer mitochondrial membrane is crossed by VDAC1–3, wide aqueous pores that pass metabolites and are the route by which cytosolic calcium reaches the intermembrane space and thus the uniporter [76,77,78]. They are counted in the permeable tier.

This is a genuinely arguable inclusion, and worth flagging as such. VDAC is not a calcium channel in the sense Cav is: it is a β-barrel with a wide pore and little discrimination, whose calcium permeability is a consequence of its general permissiveness. It is counted because it is a real and physiologically important transmembrane calcium route, and removing it costs the census three genes. Its permeability is recorded as "not applicable — a wide aqueous pore" rather than as a ratio, which is the honest form of the claim.

### 6.3 Efflux, which is not conduction

Calcium leaves the mitochondrion through NCLX, a sodium/calcium exchanger [79], and LETM1 has been reported as a calcium/proton antiporter [80]. Both are transporters and both are excluded on mechanism. LETM1 is worth naming because it sits in the catalogue's exchanger family by function while carrying none of that family's domain: this project measured its architecture directly and found the LETM1 domain and no sodium/calcium-exchanger domain at all. The family groups proteins by the job they do, and the annotation records that they are not relatives.

## 7. The TRP division: one superfamily, every answer

The TRP channels are where the difficulty of a calcium census is most concentrated. They are a single superfamily by descent, they span the whole range from $Ca^{2+}$-selective to $Ca^{2+}$-impermeable, and the boundary between those states falls *inside* families rather than between them. The founding member was found in a *Drosophila* phototransduction mutant, and the mammalian division has grown to some 28 human genes across six families [81,82]. They are also, as a group, heavily represented in inherited disease, across a range of organ systems that reflects how widely the division is deployed rather than any shared mechanism [83]. Their permeabilities were surveyed early and the survey's conclusion has held: TRP selectivity is a spectrum, and it is not predictable from family membership [12].

### 7.1 TRPV: a family split by one residue ring

TRPV1 is the capsaicin and heat receptor, the founding member of the mammalian nociceptive TRPs and a non-selective cation channel [84]. TRPV5 and TRPV6 are its close relatives and are $Ca^{2+}$-selective epithelial channels responsible for the apical step of transcellular calcium reabsorption in kidney and intestine [43,44,45,46].

This review's catalogue therefore splits TRPV into two families on measured permeability rather than on sequence similarity — a selective pair and a non-selective remainder — and records that they carry the same Pfam accessions and occupy the same length band. The split is honest and uncomfortable: it is a division of a natural sequence group by a functional measurement, and the only positive test available for

it is the projected aspartate ring of §2.4, whose discriminating power is itself something to be measured rather than assumed.

### 7.2 TRPM: $Ca^{2+}$-permeable and $Ca^{2+}$-activated in the same family

TRPM is the same problem again, and worse. TRPM2, TRPM3, TRPM6 and TRPM7 are $Ca^{2+}$-permeable; TRPM4 and TRPM5 are $Ca^{2+}$-*activated* channels that conduct monovalent cations and no calcium at all [16,17]. TRPM6 and TRPM7 in fact prefer $Mg^{2+}$ and $Zn^{2+}$ to $Ca^{2+}$, and carry a fused α-kinase domain — a channel with an enzyme bolted on, which is exactly what a one-domain-of-interest assumption misfiles.

The census excludes TRPM4 and TRPM5 and counts the rest. There is no sequence test that makes this call: the potassium filter does not fire on them, they share every Pfam accession with their $Ca^{2+}$-permeable relatives, and they sit in the same length band. The exclusion is a literature call, is recorded as one, and is precisely the kind of boundary that a census must be able to state it cannot derive.

### 7.3 The rest of the division

TRPC is the closest to the *Drosophila* founder and is the family most dependent on assembly: TRPC1 behaves differently alone than in TRPC1/TRPC4 or TRPC1/ORAI1 complexes, so its permeability is a property of the complex rather than of the gene. TRPA1 is a single human gene and a chemical irritant sensor. TRPML (§4.3) and TRPP are intracellular. PKD2 is the contested case: established in the ER and the primary cilium, disputed in the plasma membrane, and counted with the dispute recorded.

### 7.4 What the annotation does to this division

The TRP channels are the clearest demonstration in this review that domain annotation does not partition the subject the way the textbooks do. Measured against InterPro:

- `PF00520` (Ion_trans), the canonical P-loop pore model, is carried by TRPV1, TRPV5, TRPA1 and TRPM2 — and **not** by TRPC3, TRPM8, MCOLN1, MCOLN2 or PKD2. Their pores are modelled by `PF08344`, `PF18139` and `PF08016` instead. A census that enumerates P-loop channels by `PF00520` loses most of the TRP division without reporting a gap.
- `PF16519` (TRPM_tetra), a tetramerisation domain *named for* TRPM, is annotated on TRPM7 and is absent from TRPM2, TRPM8, TRPM4 and TRPM5 (measured 2026-08-20). It is carried by a minority of the family it is named after.
- `PF08016` (PKD_channel) is carried by TRPML, TRPP **and** polycystin-1 — which forms no pore on its own.

None of these is a database error. They are what a set of models built from the best-studied members of each family looks like when it is asked about the members nobody studied, and they are the reason the census behind this review runs profile-HMM and genomic sweeps in addition to domain enumeration (*pending: S2, S3, S5*).

## 8. Ligand-gated calcium entry, and a permeability not in the genome

Three unrelated superfamilies of ligand-gated channels carry physiologically important calcium currents: the ionotropic glutamate receptors, the Cys-loop nicotinic receptors, and P2X. They have nothing in common structurally, and each provides a different reason why calcium permeability resists sequence-based prediction.

### 8.1 NMDA receptors: selective, and obligately heteromeric

NMDA receptors are the most calcium-permeable of the glutamate receptors, with a fractional calcium current near 12 %, and their calcium entry is the trigger for most forms of synaptic plasticity [85,86]. They are counted in the selective tier.

They also make the sharpest case for the census counting *genes* rather than *channels*. Seven human `GRIN` genes encode NMDA receptor subunits and not one of them makes a channel alone: the receptor is an obligate heterotetramer of GluN1 with GluN2 or GluN3 subunits, and GluN3-containing receptors are far less calcium-permeable than GluN2-containing ones. Counting genes is reproducible; counting "channels" would require deciding which subunit combinations exist in which cells, which is not a genomic question. The census counts genes and says so.

### 8.2 AMPA and kainate receptors: the permeability ADAR2 decides

AMPA receptors are the clearest case in all of calcium biology of a functional property that the genome does not encode. A receptor containing an edited GluA2 subunit is calcium-impermeable; the same receptor without it is calcium-permeable. The difference is a single codon in `GRIA2`, changing a glutamine to an arginine in the pore-lining M2 loop, and the change is made post-transcriptionally by the RNA editing enzyme ADAR2 [14,87,88].

The genetics are unambiguous about which state matters. Mice lacking ADAR2 die young with seizures, and the phenotype is rescued completely by a knock-in allele encoding the edited residue directly — the enzyme's only essential substrate, in that context, is that one codon [15].

For a sequence-based census this is a hard limit rather than a difficulty. The genomic `GRIA2` sequence encodes the calcium-permeable form; editing is close to complete in adult brain; and no method operating on genomes or proteomes can recover the distinction. The catalogue therefore places AMPA and kainate receptors in a **conditional** tier, records the mechanism, and records that there is no test. A census that quoted a single number for "calcium-permeable human genes" without that tier would be asserting something about nine genes that nothing in the data supports.

### 8.3 Nicotinic receptors: a boundary that splits sisters

The Cys-loop nicotinic acetylcholine receptors span an order of magnitude in calcium permeability. The α7 homopentamer reaches P_Ca/P_Na near 10 and carries calcium entry sufficient to trigger downstream signalling; the muscle-type receptor is near 0.1–1 [13]. The catalogue counts α7, α9 and α10 and excludes the rest.

This is the census's most uncomfortable boundary, and it is worth stating plainly rather than smoothing over. It divides a single natural sequence family by a measured ratio, placing sister subunits on opposite sides. If a tree of the nicotinic subunits does not separate them, the split is honest and awkward; if it does, it is a real clade. Which of those is true is a result to be produced, not assumed (*pending: S7*).

### 8.4 P2X: a third fold, and a pore that changes size

P2X receptors are ATP-gated trimers with two transmembrane helices per subunit and no relationship to either of the families above; the zebrafish P2X4 structure established the fold [89,90]. The seven human paralogues span roughly an order of magnitude in calcium permeability, with P2X1 and P2X4 highest [91].

P2X7 adds a further complication of the kind that defeats any static description: on prolonged activation its pore dilates and begins to pass molecules up to about 900 Da. A permeability measured at one moment of activation is not the permeability at another, which is one more reason the catalogue stores these ratios as strings with their conditions attached rather than as numbers.

## 9. Mechanically activated calcium entry

Mechanosensitivity has been invented many times, and three of its solutions carry calcium in a way that puts them in this census. None is related to any other, and none is related to the voltage-gated channels.

### 9.1 Piezo

PIEZO1 and PIEZO2 were identified in a screen for the mechanically activated current of a neuroblastoma line and are the mechanotransducers of touch, proprioception, vascular development and red-cell volume regulation [92,93]. They are very large — over 2,500 residues — and their structures show a three-bladed propeller whose curved arms deform the membrane and are thought to gate the central pore by flattening [94].

They are non-selective cation channels with a modest calcium preference, and they are counted in the permeable tier. Architecturally they are isolated: their own superfamily, their own fold, and no alignable relationship to anything else counted here.

### 9.2 OSCA / TMEM63

The OSCA channels were found in *Arabidopsis* as the hyperosmolality-gated calcium-permeable conductance, in a screen for plants that failed to raise cytosolic calcium in response to osmotic stress [95]. Their

animal relatives are the TMEM63 proteins, and the family is now understood as a widespread mechanosensitive channel class.

The catalogue counts the family and records an asymmetry in the evidence: the plant OSCA1 is well characterised as calcium-permeable, while divalent permeability in the mammalian TMEM63 proteins is less thoroughly measured. Mechanosensitivity is not in doubt; the calcium claim is stronger on one side of the family than the other, and the record says so.

### 9.3 Plant MCA

MCA1 and MCA2 are *Arabidopsis* plasma-membrane proteins required for mechanically induced calcium influx in roots, identified by complementation of a yeast mutant [96]. They are small, carry a PLAC8 domain, and have no animal counterpart in this census. They are one of several demonstrations in §10 that plant calcium entry does not run through plant versions of animal channels.

## 10. Plants, fungi and viruses

A census restricted to human genes cannot make an evolutionary claim, and the non-animal calcium channels are emphatically not the animal ones renamed. Three facts organise this section, and each of them is a constraint on any statement about how calcium signalling evolved [97,98,99].

### 10.1 Plants have no Cav

There is no plant homologue of the voltage-gated calcium channel family. This is not a gap in annotation; it is an absence, and plants nonetheless perform voltage- and ligand-dependent calcium signalling through two families that are relatives of animal channels doing other jobs:

- **The glutamate receptor-like channels (GLR)** are homologues of the animal ionotropic glutamate receptors. They carry the wound-induced, long-distance calcium signal that propagates from a damaged leaf to undamaged ones, and the *glr3.3 glr3.6* double mutant loses it [100,101]. Note the form of that evidence: it is genetic evidence for a pathway, not a measured permeability ratio, and the catalogue records which kind of evidence it has.
- **The cyclic-nucleotide-gated channels (CNGC)** are relatives of the animal CNG and HCN channels, and several members are individually characterised as calcium-permeable.

Plants also have a two-pore channel, TPC1, in the vacuolar membrane — the slow vacuolar current, and one of the better-characterised plant calcium conductances.

The GLR case is also a hazard, and this project records it as one. The plant channels carry all three of the domains the animal iGluRs carry — measured 2026-08-20 on *Arabidopsis* GLR3.3 and GLR3.4 — so nothing in the domain architecture distinguishes a plant glutamate receptor-like channel from an animal AMPA, kainate or NMDA receptor. A domain-based census that assumed otherwise would misassign them silently.

### 10.2 Fungi kept Cav and lost MCU

Fungi retain a voltage-gated calcium channel homologue, CCH1, and have lost the mitochondrial uniporter. *Saccharomyces cerevisiae* accordingly performs no mitochondrial calcium uptake — the fact that made the uniporter invisible to yeast genetics for decades and that, inverted into a phylogenetic-profiling screen, is what finally identified MICU1 and MCU (§6.1) [38,40].

Whether that loss is a single event or several, and how it maps onto the fungal tree, is a question for ancestral reconstruction across the lineage panel rather than one this review can settle (*pending: S10*).

### 10.3 Algae, and the origin questions

The photosynthetic eukaryotes are where the origin questions are decided, because they retain combinations that neither plants nor animals kept. Surveys of calcium channel repertoires across algal genomes find a distribution that does not match either the plant or the animal pattern, and they are the reason the lineage panel behind this census extends well beyond the standard model organisms [102].

### 10.4 Viral viroporins

Several viruses encode small membrane proteins that form calcium-permeable pores, and they are counted here because they meet the definition and because excluding them would be a decision about taxonomy rather than about mechanism. Rotavirus NSP4 is the best case: a genuine ER calcium-release channel, and the virus's enterotoxin. At roughly 175 residues it is the smallest protein in this review's catalogue — against RYR1's 5,038, a thirty-fold span within one census. The SARS-CoV-2 E and ORF3a proteins are also catalogued here.

The viroporins are marked as a superfamily that cannot be aligned across, and they are one of the cases where the project's phylogeny protocol refuses to build a tree rather than producing one from an alignment that does not exist.

## 11. What the word "calcium" costs

The largest source of error in a calcium census is not a difficult phylogenetic call. It is the word. "Calcium" attaches to proteins that are gated by calcium, that pump calcium, that exchange calcium, that bind calcium, that sense calcium, and that assemble with calcium channels — and to almost none of these does it attach because they conduct calcium down a gradient through a pore. This section works through the excluded set, because in this project the excluded set is a deliverable rather than housekeeping: the catalogue holds **245 human genes and counts 98**, and the 147 it does not count are what a search for "calcium channel" returns.

### 11.1 Gated by calcium, conducting something else

The $Ca^{2+}$-*activated* channels are the most-cited contaminant.

| family | genes | conducts |
|---|---|---|
| BK | KCNMA1 | $K^+$ |
| SK / IK | KCNN1–KCNN4 | $K^+$ |
| TMEM16A/B | ANO1, ANO2 | $Cl^-$ |
| TRPM4/5 | TRPM4, TRPM5 | monovalent cations |

They are opened by cytosolic calcium and they carry no calcium current. They appear in IUPHAR class names, in review titles and in gene-set annotations with "calcium" in them, and a keyword-driven list counts every one.

For the potassium channels a positive test exists and is decisive: the T-x-G-Y-G filter says the protein conducts $K^+$, without reference to a name or a symbol. For TRPM4 and TRPM5 no such test exists (§7.2), and the exclusion is recorded as a literature call.

### 11.2 Transporters

Nineteen human genes move calcium across membranes by mechanisms that are not conduction:

- **P-type ATPases** — SERCA (ATP2A1–ATP2A3), PMCA (ATP2B1–ATP2B4) and SPCA (ATP2C1, ATP2C2) — hydrolyse ATP and move calcium against its gradient through an alternating-access cycle. SERCA is among the best-characterised membrane proteins in structural biology [18,19].
- **$Na^+/Ca^{2+}$ exchangers** — NCX (SLC8A1–SLC8A3), NCKX (SLC24A1– SLC24A5) and NCLX (SLC8B1) — couple calcium movement to the sodium gradient [20,79].

These are excluded on mechanism, not on importance; several move more calcium per cell than most census families do. The distinction being drawn is between a pore that is open at both ends at once and a cycle that is not.

### 11.3 Sensors and buffers, which conduct nothing

Calmodulin, calbindin, parvalbumin, the S100 proteins and the calcium-sensing receptor CASR all bind or sense calcium and none conducts it [103,104]. They are catalogued as non-channel homologues, and they matter to a census for a specific reason: they are what an EF-hand-driven or annotation-driven search returns.

The positive test here is the EF-hand itself, used as a *detector of decoys* rather than as a veto (§5.2).

### 11.4 The subunits

Forty human genes encode proteins that assemble with calcium channels and form no pore: the Cav β, α2δ and γ subunits, the CatSper accessories, STIM1 and STIM2, the uniporter regulators MICU1–3, EMRE and MCUR1, the release regulators calsequestrin, FKBP12/12.6, triadin and junctin, and polycystin-1.

Sixteen of them carry a `CACN` symbol prefix, and seven of those sixteen — CACNG2 to CACNG8 — are AMPA receptor subunits with no role in voltage-gated calcium channels at all (§3.3). Any census assembled from symbol prefixes counts all sixteen, and any census assembled from protein-complex membership counts all forty.

### 11.5 The rule, and why it is a rule

Across every case above, the catalogue's requirement is the same: **a family is excluded by a positive test, never by a caveat and never by a symbol.** Not "this is not a Cav because the symbol says `SCN5A`", but "the filter reads DEKA". Not "TRPM4 is probably not in scope", but either a test or an explicit statement that no test exists and the call is a literature one.

The reason is that a census whose exclusions are unexamined is unfalsifiable. Listing 98 genes and moving on would look tidier and would give a reader nothing to check. Listing 245 with a recorded reason for each of the 147 exclusions makes every one of them arguable — and each is designed to be reversible by editing a single field and recounting. Moving the connexin hemichannels into scope adds 16 genes; removing TPC costs 2; removing VDAC costs 3; tightening the nicotinic boundary costs 3; dropping the conditional tier entirely costs 9.

## 12. What the databases record

Everything in this section was measured against live services rather than recalled, by `scripts/s0_catalogue_verify.py`, `scripts/s0_rule_coverage.py` and `scripts/s0_anchors.py`, and is committed as tables under `results/s0_baseline/`. The purpose is not to criticise the resources — they are the reason a census is possible at all [105,106,107] — but to establish what a census built on them can and cannot claim.

### 12.1 The catalogue's own claims, re-derived

The catalogue is a file full of assertions: this Pfam accession exists and means that; this gene symbol resolves to that protein in that organism; this family's members carry these domains; this taxon id is that species. Written by hand and checked by nothing, such a file is a liability. Re-deriving all of it against live databases produced the following state:

| checked | n | result |
|---|---|---|
| Pfam accessions | 87 | all resolve; every short name matches |
| exemplar proteins | 102 | all resolve to a declared accession |
| taxon ids | 53 | all resolve |
| projection anchors | 3 | all read what the catalogue claims |
| declared architectures | 102 | 30 show an absence, all subfamily-level |

That table is the end state, and it is worth being explicit that it was not the starting one. The verification found and corrected a series of errors, and they are more interesting than the clean result.

### 12.2 Errors that a name check would not have caught

**Two Pfam families share a description.** PF00702 and PF08282 are both called "haloacid dehalogenase-like hydrolase". The catalogue declared PF00702 for the P-type $Ca^{2+}$-ATPases; InterPro annotates PF08282 on both SERCA and PMCA, and neither pump carries PF00702. The accession existed and its name matched what had been written, so every check short of asking whether the family's own members carry it would have passed.

**A domain named for a family is carried by a minority of it.** PF16519 (TRPM_tetra) is annotated on TRPM7 and is absent from TRPM2, TRPM8, TRPM4 and TRPM5.

**A domain the hazard registry had already flagged was still being used to name a family.** PF00027 (cNMP_binding) spans the animal CNG channels, HCN and the plant CNGCs — recorded as a hazard, and simultaneously declared as a family-level signature of animal CNG. The contradiction was invisible while a second requirement masked it. Correcting an unrelated over-claim removed that mask, and the generated rule immediately named the plant CNGC channels as animal CNG channels. The lesson generalises: a correction can create a defect, and only a coverage test that re-derives every rule from the catalogue after every edit will catch it.

**A "family" can have no shared domain at all.** Several catalogued groups are defined by the job their members do rather than by descent, and the union of their members' domains had been declared as though every member carried all of them. Measured, they do not:

| group | members carry |
|---|---|
| Cav auxiliary subunits | β: guanylate kinase; α2δ: VWA; γ: claudin |
| uniporter regulators | MICU1: EF-hands; EMRE: PF10161 alone |
| release regulators | CASQ2: calsequestrin; FKBP1A: FKBP |
| connexin / pannexin | GJA1: connexin; PANX1: innexin |

These are not annotation failures. They are the catalogue asserting a shared architecture where the biology has convergence or mere co-membership, and the correction is to record the level at which each signature is actually diagnostic.

### 12.3 A gene symbol is not an identifier

The reference panel that the classifier scores against is built by resolving gene symbols to UniProt entries. Symbols are recycled, and UniProt records synonyms: GRIK5 (Q16478, 980 aa) carries GRIK2 in its synonym list, so a symbol query for GRIK2 returned two reviewed human entries and a longest-wins tiebreak selected the wrong protein. The panel had been carrying GRIK5's sequence under the label Hs_GRIK2.

This is worth stating in a review because it is a failure mode of the *tooling*, not of the databases, and because the project's own rule — never consult a gene symbol when classifying — had been obeyed by the classifier and quietly broken by the code that built its references. Every exemplar in the catalogue now declares an accession explicitly, and the panel no longer depends on what a symbol lookup returns on a given day.

A related case is benign and equally instructive: *Arabidopsis* OSCA1.1 resolves only as a synonym of UniProt's OSCA1. Reporting that as "the accession names the wrong gene" would be false, so the check distinguishes a synonym match from a mismatch.

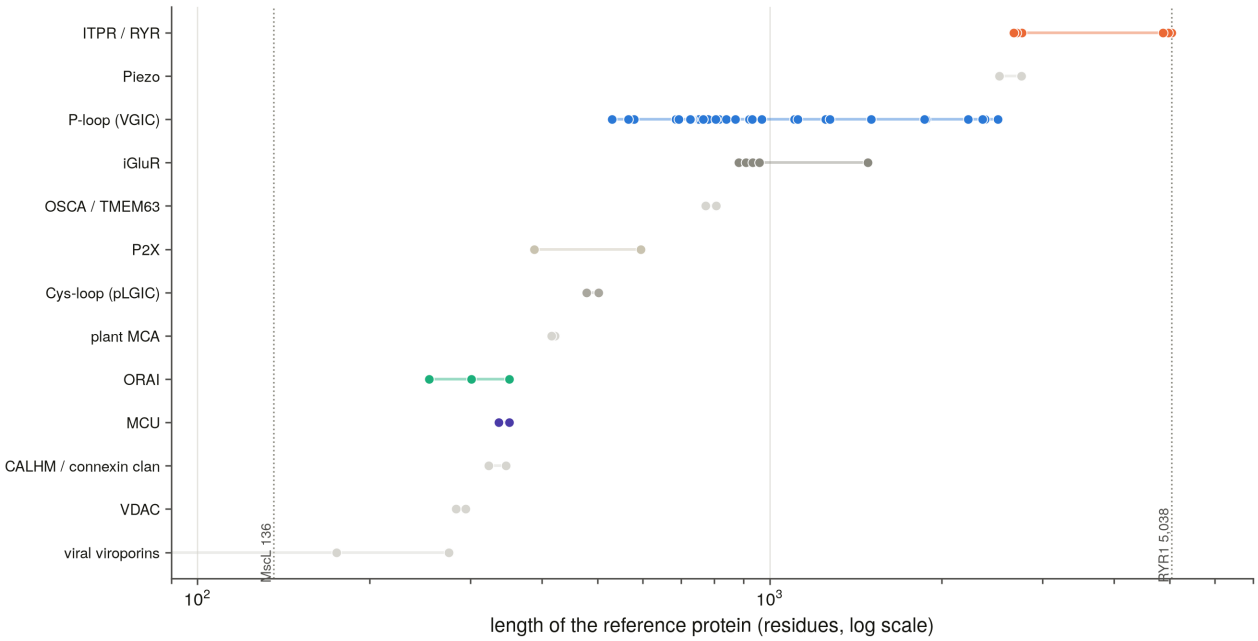



**Figure 4 | Thirty-fold, within one census.** Observed length range per superfamily, from the catalogue's exemplars. The subject spans rotavirus NSP4 at roughly 175 residues to RYR1 at 5,038 — a range no single size filter can accommodate, and the reason the catalogue declares a size band per family rather than one for the census.

### 12.4 Domain architecture is not a family call

The strongest quantitative statement this project can currently make about annotation is a coverage measurement: feed every family its own declared architecture back through the classifier's architecture tier, and ask whether the family is named.

**17 of 29 census families are named. Twelve are not.** The twelve are not a random sample — they are, almost exactly, the families the hazard registry already identifies as architecturally indistinguishable from a sibling:

- the T-type calcium channels and CatSper, identical in architecture to the sodium channels (§3.2);
- both TRPV families, and TRPA, within the TRP division (§7.1);
- the nicotinic α7 group against the rest of its own family (§8.3);
- all four glutamate-receptor families — AMPA, kainate, NMDA and the plant GLRs — against one another (§8, §10.1);
- the two-pore channels and the plant CNGCs.




**Figure 8 | The architecture traps, to scale.** The families that share an architecture with something they are not, drawn with their declared domains in proportion. Cav3 against Nav is the extreme case: identical domain content in identical copy number, separated by the four residues of Figure 3.

"Domain architecture is not a family call" is a claim this review can attach a number to, and the number is 12 of 29. What those twelve need is the motif tier or the reference tier, and how far each gets there is measured by the classifier benchmark rather than assumed (*pending: S1*).

### 12.5 Shared signatures, and the census boundary

Eighteen domain signatures in the catalogue are carried by more than one family, and the interesting subset is those that cross the census boundary — a domain that is evidence for a calcium-channel family and is also carried by something the census does not count. `PF00520` in four copies spans Cav, Nav, NALCN and

CatSper. PF02815 (MIR) spans ITPR, RYR and the O-mannosyltransferase POMT1. PF08016 spans TRPML, TRPP and polycystin-1. PF00027 spans CNG, HCN and plant CNGC.

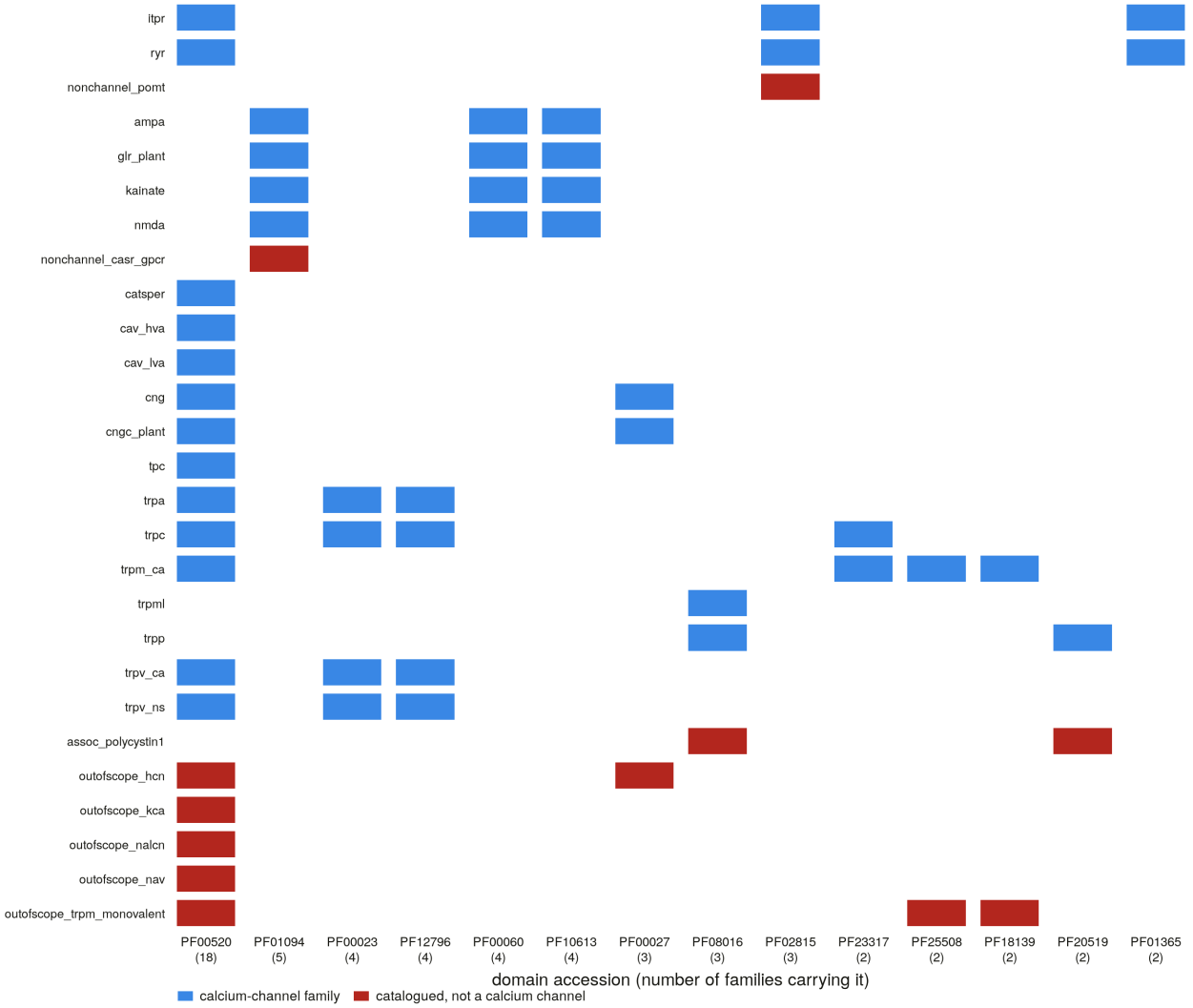



**Figure 5 | The domains that cross the boundary.** Every signature carried by more than one catalogued family, with the families that carry it, and with those crossing the census boundary marked — rendered from `signature_sharing.tsv`. A marked row is a domain that is evidence for a calcium-channel family and is also carried by something the census does not count: a sodium channel, a $Ca^{2+}$-activated channel, a pump, an auxiliary subunit or a sensor.

Each of these is a place where architecture alone cannot decide membership, and each is why the classification behind this review assigns by positive test and records which tier of evidence decided.

## 13. How many times has a calcium-selective pore evolved?

### 13.1 The question, and why it has no single answer yet

Calcium signalling is old. It is present in bacteria, and the core logic — keep cytosolic calcium low, then use its controlled entry as a signal — is shared across the tree, which suggests it was established very early [97,98]. But the *channels* are not one story, and the question in this section's title is deliberately awkward because it cannot be answered inside any single tree.

The honest current answer is **at least four, in unrelated folds**: the EEEE ring of the voltage-gated channels, ORAI's E106, the uniporter's DIME motif, and the TRPV5/6 aspartate ring. These four share no alignable position with one another, so "four" is a count of unconnected components in a structural network, not a count of clades.

That framing carries an important caveat, and this review states it rather than hiding it. "Unrelated" may be a fact about the proteins or a fact about the detection limit of the methods used to compare them.

Distinguishing those two possibilities requires structural comparison — all-against-all structural alignment across the fold network, rather than sequence similarity — and it is an open result here (*pending: S12, S18*).

### 13.2 What is settled

**The P-loop pore module is ancient and prokaryotic.** The voltage-gated superfamily descends from a prokaryotic ancestor, and bacterial channels remain the structural prototypes and the outgroups that root the trees [53].

**The four-repeat architecture arose by internal duplication.** Cav and Nav are single polypeptides of four homologous repeats, built from a six-transmembrane ancestor by two rounds of internal duplication. The two-repeat two-pore channels are the intermediate the story predicts. Their placement is a result this project's pore-module trees are meant to produce rather than an assumption they start from (*pending: S7, S8, S11*).

**Cav is animal, and does not exist in plants.** The four subunits of the voltage-gated calcium channel complex have distinct evolutionary histories, with ancient roots, increasing complexity and multiple independent losses across lineages [108]. Plants have no Cav homologue at all; fungi kept one and lost the uniporter (§10).

**Repertoires differ profoundly across eukaryotes.** Surveys of algal and other photosynthetic-eukaryote genomes find calcium channel complements that match neither the plant nor the animal pattern [102].

### 13.3 What is open, and how this project intends to close it

Four questions are live, and each is stated here with the evidence that would settle it:

1. **Is Cav's absence from plants a loss or a never-had?** Answerable only against a declared lineage panel with a genomic sweep behind it, so that an absence is a claim about a search space rather than about the limits of a proteome download (*pending: S5, S10*).
2. **Is MCU loss in fungi one event or several?** Ancestral reconstruction across the panel (*pending: S10*).
3. **Is the selectivity filter congruent with the pore-module tree?** That is, has calcium selectivity been gained and lost *within* families, as the TRPV and TRPM splits suggest? This is the question that would turn the TRP division from a taxonomic problem into an evolutionary result (*pending: S9*).
4. **Are the four folds genuinely unrelated?** (*pending: S12, S18*.)

### 13.4 A methodological point about trees

Because superfamilies are defined by descent and not by ion, a pore-module tree of the P-loop superfamily contains the voltage-gated sodium channels and the $Ca^{2+}$-activated potassium channels alongside the calcium channels. There is a standing temptation to prune them and present "the calcium channel tree". Doing so would hide the non-monophyly that is the single most important fact about the subject, and would remove the outgroup that makes the tree rootable.

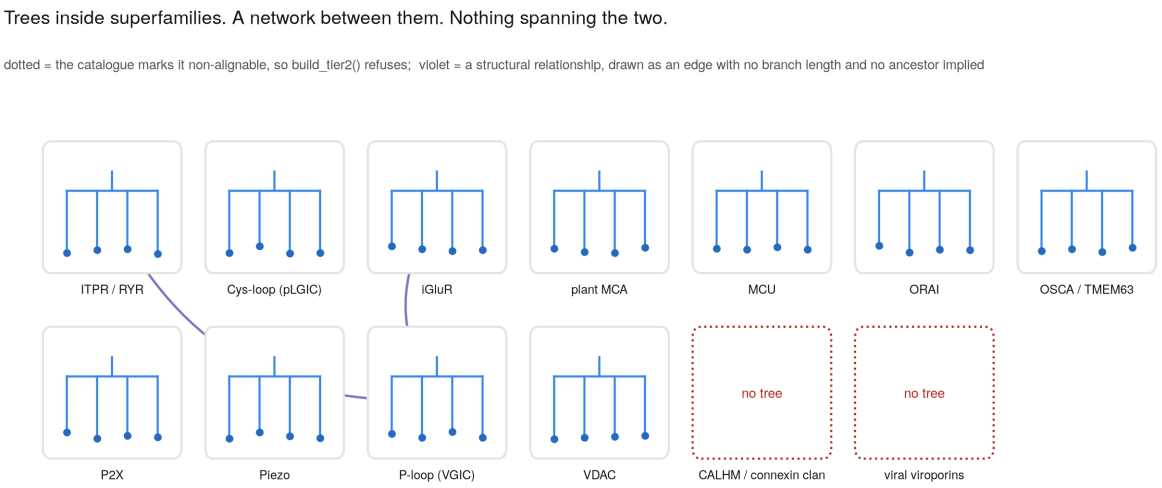



**Figure 7 | A forest, and the trees that are refused.** The phylogenetic units this project will build: tier-1 trees within families, tier-2 pore-module trees across alignable superfamilies including their non-calcium members, and the superfamilies for which no tree is built at all because no alignment across them exists. The figure is a schematic of a rule rather than a result — it displays which units exist, not any inference from them.

The protocol behind this review therefore builds a forest — trees within families and within alignable superfamilies, keeping every member — and refuses outright to build a tree across a superfamily marked non-alignable. The refusal is implemented in code rather than left to discipline: a request for a tree spanning the CALHM large pores or the viral viroporins raises an error and reports why.

## 14. Disease, and the calcium diseases that are not channelopathies

Calcium channels are among the most clinically consequential ion channel families, both as disease genes and as drug targets [109,110]. This section is organised around a distinction the census makes possible: between diseases caused by calcium channels and diseases of calcium signalling caused by something that is not one.

### 14.1 Channelopathies of the census families

| gene | disease |
|---|---|
| `CACNA1S` | hypokalaemic periodic paralysis; malignant hyperthermia susceptibility |
| `CACNA1C` | Timothy syndrome; Brugada-pattern arrhythmia |
| `CACNA1A` | familial hemiplegic migraine; episodic ataxia type 2; SCA6 |
| `CACNA1F` | congenital stationary night blindness |
| `RYR1` | malignant hyperthermia; central core disease |
| `RYR2` | catecholaminergic polymorphic ventricular tachycardia (CPVT) |
| `ORAI1` | severe combined immunodeficiency with anhidrotic ectodermal dysplasia |
| `MCOLN1` | mucolipidosis type IV |
| `PKD2` | autosomal dominant polycystic kidney disease |
| `TRPV6`, `CNGA1/CNGB1` | transient neonatal hyperparathyroidism; retinitis pigmentosa |

Two of these are worth drawing out. `ORAI1` is the case where a human immunodeficiency pedigree and an

unbiased screen converged on the same gene, which is much of why the CRAC channel assignment was accepted so quickly [35]. CACNA1A is the case where three clinically distinct neurological syndromes — a migraine, an episodic ataxia and a spinocerebellar ataxia — map to one calcium channel gene.

### 14.2 The diseases of calcium signalling that are not channel diseases

The census's excluded set is not clinically inert. Several textbook "calcium-signalling diseases" are caused by proteins that conduct no calcium:

- CASQ2 **causes CPVT** — the same disease as RYR2 mutations, produced through a luminal calcium buffer rather than through the pore it regulates. Identical phenotype, different kind of protein.
- CALM1**–**CALM3 **mutations cause long-QT syndrome and CPVT.** Calmodulin is an EF-hand sensor and conducts nothing.
- TMEM38B **causes osteogenesis imperfecta** — a calcium-release disease caused by a *potassium* channel, TRIC-B, which supplies the counter-current for release [70].
- ATP2A1 **causes Brody myopathy** and ATP2C1 **Hailey–Hailey disease** — both pumps [19].

A gene list assembled from calcium-signalling disease phenotypes would therefore contain buffers, sensors, pumps and a potassium channel, and would still be missing most of the census. This is the clearest practical argument for defining membership by mechanism and reporting the exclusions.

### 14.3 Pharmacology, and a drug target that is not a channel

The voltage-gated calcium channels are a major drug class: dihydropyridines, phenylalkylamines and benzothiazepines acting on Cav1 in hypertension and angina, ethosuximide on Cav3 in absence epilepsy, and ziconotide — a ω-conotoxin — on Cav2.2 for intractable pain [55,111,112].

The observation this review wants to record is a different one. **Gabapentin and pregabalin, among the most-prescribed drugs acting anywhere on this subject, bind α2δ-1 — an auxiliary subunit that forms no pore and is not in the census** [21,56]. A pharmacology-driven definition of "calcium channel" would necessarily include it, and would be drawing the boundary around the drug rather than around the mechanism. The catalogue keeps α2δ-1 as an excluded, catalogued, clinically central non-channel, which is exactly the kind of entry the excluded set exists to hold.

## 15. What is unresolved

This section separates three kinds of uncertainty that are easy to conflate: identities that are genuinely disputed in the literature, quantities this project has not yet measured, and limits that no sequence-based method can pass.

### 15.1 Contested identities

Recorded in the catalogue as contested rather than resolved by omission:

- **TPC1/TPC2** — named as calcium channels and reported as NAADP-gated calcium release from acidic stores [66]; direct endolysosomal patch clamp finds a PI(3,5)$P_2$-activated, $Na^+$-selective current [67]. Counted, in the release tier, with the dispute recorded.
- **PKD2** — established in the ER and the primary cilium, disputed in the plasma membrane.
- **CALHM4–6** — curated into the family by homology; only CALHM1 and CALHM3 have direct functional data [113,114].
- **TMEM63A–C** — mechanosensitive for certain; divalent permeability less thoroughly established than in the plant OSCA1.
- **Connexin hemichannels** — excluded on the ground that the physiological state is the docked junction, which connects two cytosols rather than crossing a gradient. Undocked hemichannels do pass calcium, and the dissent is recorded rather than resolved by omission. CALHM1 shares structural features with connexins, pannexins and innexins, which is why the two sit adjacent in the catalogue [115].

### 15.2 The weakest claims in this review

Two classes of statement here rest on less than the rest.

**Every permeability ratio is a literature value, and no database holds one.** The tiers that organise this entire review — selective, permeable, conditional, release — are boundaries drawn on P_Ca/P_Na values that were recorded by hand from the literature. Nothing in the verification described in §12 checks them, because no database contains them. Tracing each one to a resolved source with its method attached is a separate audit, and until it runs, the tier boundaries rest on claims that nothing has checked (*pending: S1b*). This is stated here as prominently as it is because a reader should discount the tier assignments accordingly.

**Permeability ratios are not single numbers.** They are measured in a particular cell, at a particular divalent concentration, and often per subunit combination. The catalogue stores them as strings — `">1000"`, `"~1-10"`, `"~0.5-4 (P2X1 and P2X4 highest)"` — because turning a range measured in one cell type into a float would invent a precision nobody measured, and the census boundary would then rest on it.

### 15.3 Limits no sequence method can pass

Three things are not recoverable from a genome, and this project does not attempt them:

1. **RNA editing.** Whether an AMPA receptor conducts calcium is set by ADAR2 on one `GRIA2` codon (§8.2).
2. **Subunit composition.** NMDA receptor permeability depends on whether GluN2 or GluN3 is present; TRPC1 behaves differently alone and in complexes.
3. **Use-dependent pore dilation.** TRPV1 and P2X7 change their permeation properties during prolonged activation (§8.4).

A census that reported a permeability for every gene would be asserting things about these cases that no data supports. The catalogue therefore **never predicts a permeability** — the classifier returns a family, a superfamily, a status and an audit trail, and no calcium claim at all.

### 15.4 Open measurements

Finally, the results this review is waiting on, each attached to the task that produces it:

| question | status |
|---|---|
| Per-family recall and per-hazard specificity of the classifier | *pending: S1* |
| Does the TRPV aspartate anchor actually separate TRPV5/6 from TRPV1–4? | *pending: S1* |
| Every P_Ca/P_Na traced to a source with its method | *pending: S1b* |
| What domain search alone would have missed, per superfamily | *pending: S15* |
| CatSper's filter, read by joint projection across four subunits | *pending: S9* |
| Is the filter congruent with the pore-module tree? | *pending: S9* |
| Do the nicotinic α7/α9/α10 subunits form a clade? | *pending: S7* |
| How many times has a $Ca^{2+}$-selective pore evolved? | *pending: S12, S18* |
| Is Cav's absence from plants a loss or a never-had? | *pending: S5, S10* |

## 16. Methods

This review is a component of a project that produces a census, a classification and a phylogeny, and its measured statements come from that project's committed outputs. This section states how they were produced and what the reader is entitled to conclude from them.

### 16.1 The subject is defined in one place

Every family, signature, size band, permeability, exemplar and hazard is declared in a single catalogue module, and nothing else in the codebase hard-codes a gene name, a Pfam accession or a size band. Adding a family is an edit to one file plus, at most, one hazard row. The catalogue currently holds 49

families in 17 superfamilies: 29 census families covering 98 human genes, and 20 control families that exist so they can be excluded, covering the other 147.

Every assertion carries a provenance — re-derived from a live database, taken from the literature, curated by hand and not yet checked, or explicitly open. Curated-but-unchecked is a legitimate state; silently wrong is not. As of this build, no signature in the catalogue remains in the unchecked state.

### 16.2 The census is reported in tiers

Membership is a measured permeability, so the census is reported as four tiers with the evidence for each attached, and a flat total is never quoted without them:

| tier | criterion | families | human genes |
|---|---|---|---|
| selective | pore discriminates for $Ca^{2+}$, roughly P_Ca/P_Na > 10 | 7 | 28 |
| permeable | non-selective, physiologically important $Ca^{2+}$ current | 15 | 50 |
| conditional | permeability set by editing, splicing or subunit composition | 2 | 9 |
| release | releases $Ca^{2+}$ from an intracellular store | 5 | 11 |

Every family carries a permeability record whose ratio is stored as a string, together with the method that produced it and its provenance; a census family whose method field is empty fails validation.

### 16.3 Classification is by positive test, with an audit trail

A protein is classified by three tiers of evidence, in precedence order:

1. **Hazard rules** — hand-written tests for recorded ways of classifying wrongly, each naming the discriminating observation.
2. **Motifs** — three reference-free regular expressions (the potassium filter, the uniporter's DIME motif, the canonical EF-hand loop) and three projected anchors (the four-repeat locus from Nav1.5, the TRPV aspartate from TRPV6, the ORAI glutamate from ORAI1).
3. **Reference identity** — alignment against a panel of catalogued exemplars, scored over mutually covered columns only, with a required margin between the best and second-best family and a coverage floor.
4. **Domain architecture**, last, and never on its own where a hazard applies.

Ties are reported as ambiguity rather than resolved by a coin toss. The gene symbol is carried through the whole pipeline and never consulted. **No permeability is ever predicted.**

Every projected anchor is verified against its own reference before use, and a non-validating anchor refuses to project rather than returning a guess. The TRPV6 case in §2.4 is why.

### 16.4 The phylogeny is a forest, and sometimes a refusal

Three tiers, and no fourth:

- **Tier 1** — full-length trees within a family, rooted on the family's declared outgroup.
- **Tier 2** — pore-module trees across an alignable superfamily, including every member whether or not it conducts calcium.
- **Tier 3** — a structural network between folds, with no branch lengths and no support values. An unmeasured pair is recorded as unmeasured, never as unrelated.

A request for a tree spanning a superfamily the catalogue marks non-alignable raises an error and reports the reason. There is no tier that builds one tree of all calcium channels, because no such alignment exists.

Alignments use MAFFT and trimAl; trees use IQ-TREE 2 with ModelFinder and ultrafast bootstrap; structural comparison uses Foldseek against AlphaFold models [116,117,118,119,120,121]; profile searches use HMMER [122]. A missing tool produces a missing result and a note, never a silently weaker method.

### 16.5 Reproducibility and the standing of the citations

Every analysis script renders its report purely from its own committed tables, so a report cannot drift from the data it describes. Verification scripts report mismatches and never edit the catalogue: a script that rewrites its own inputs to make its checks pass is not a check.

The bibliography of this review is machine-generated. Each source title is resolved against Europe PMC, and a record is accepted only if the returned title is at least 90 % similar to the one requested, so a misremembered paper fails loudly rather than becoming a plausible entry. **A cited key with no verified reference is a build error.** Every citation in this document is therefore one that a live literature service returned, not one that anybody recalled.

### 16.6 What this review does not establish

The measured statements here are the catalogue-verification results of §12. They establish that the catalogue's claims about accessions, exemplars, architectures, anchors and taxa are correct as of the build date. They do **not** establish that the catalogue is complete, that its family assignments are right, or that its permeability values are accurate — those are the census, the classifier benchmark and the permeability audit respectively, and each is marked in the text where it is relied on.

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